Practical 1

a) Create a web application using servlet to display a welcome message.

index.html

```
<!DOCTYPE html>
<html>
   <head>
      <title>TODO supply a title</title>
      <meta charset="UTF-8">
      <meta name="viewport" content="width=device-width, initial-scale=1.0">
   </head>
   <body>
      <form method="get" action="WelcomeServlet">
         <h1>A Welcome Web Application</h1>
         Enter your Name:<input type="text" name="txtName">
         <input type="submit">
      </form>
   </body>
</html>
```

WelcomeServlet.java

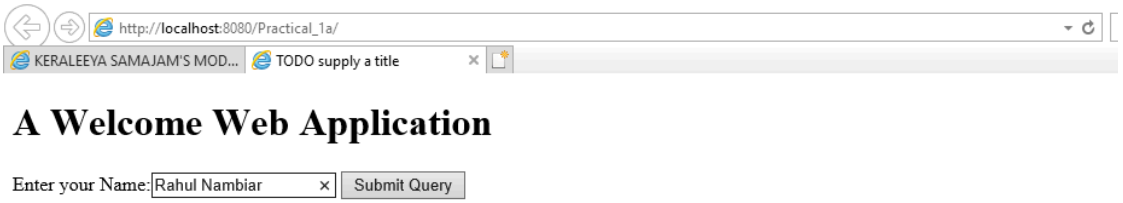
```
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

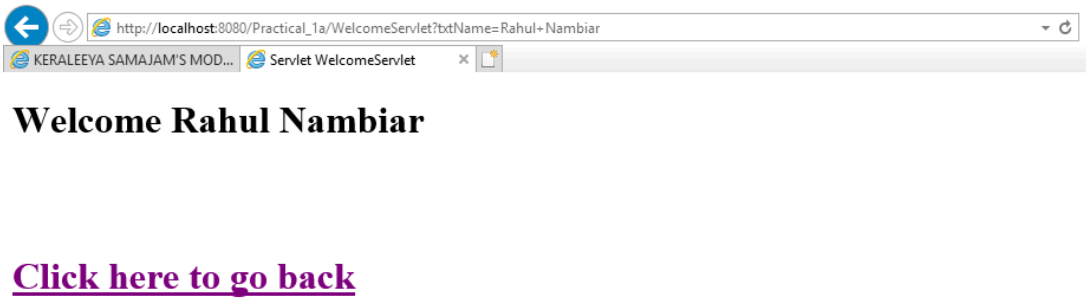
public class WelcomeServlet extends HttpServlet {
protected void processRequest(HttpServletRequest request, HttpServletResponse
response)
         throws ServletException, IOException {
      response.setContentType("text/html;charset=UTF-8");
      try (PrintWriter out = response.getWriter()) {

         out.println("<!DOCTYPE html>");
         out.println("<html>");
         out.println("<head>");
         out.println("<title>Servlet WelcomeServlet</title>");
         out.println("</head>");
         out.println("<body>");
```

```
            out.println("<h1>Welcome " + request.getParameter("txtName")+"<h1>");
            out.println("<br><br><a href='index.html'>Click here to go back </a>");

            out.println("</body>");
            out.println("</html>");
        }
    }
}
```





b) Create a servlet for a login page. If the username and password are correct, then it says message "Welcome " else a message "login failed".

index.html

```
<!DOCTYPE html>
<html>
   <head>
      <title>Login Page</title>
      <meta charset="UTF-8">
      <meta name="viewport" content="width=device-width, initial-scale=1.0">
   </head>
   <body>
      <form method="get" action="LoginServlet">
         Enter User ID:<input type="text" name="txtId"> <br>
         Enter Password:<input type="password" name="txtPass"><br>
         <input type="reset">
         <input type="submit" value="Click to Login">
      </form>
   </body>
</html>
```

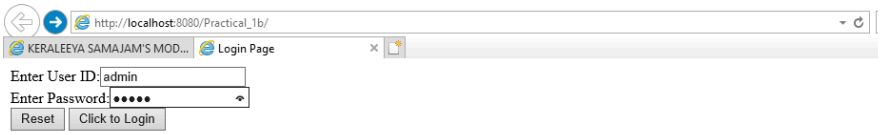
LoginServlet.java

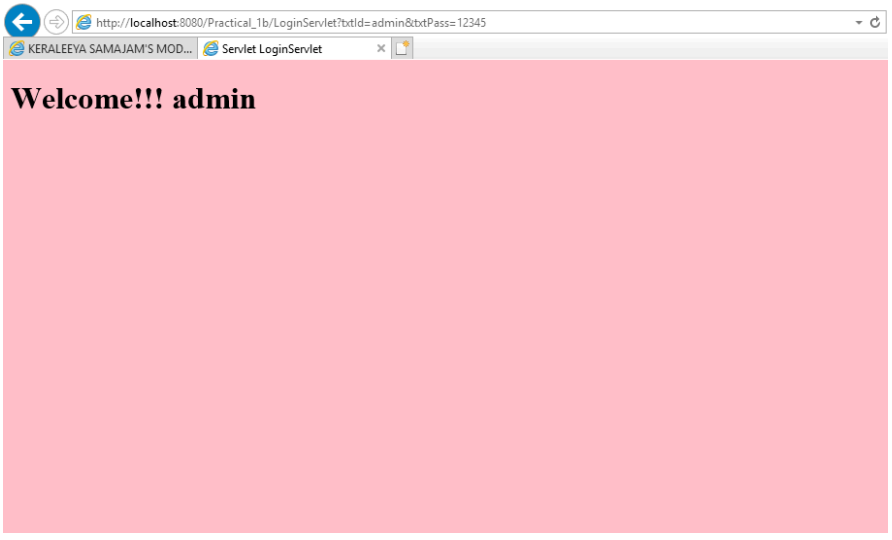
```
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

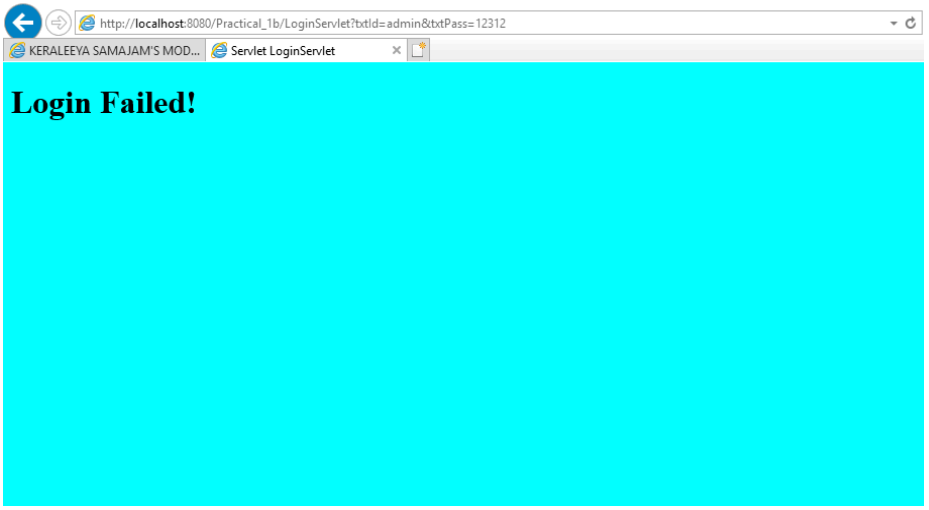
public class LoginServlet extends HttpServlet {
protected void processRequest(HttpServletRequest request, HttpServletResponse
response)
         throws ServletException, IOException {
      response.setContentType("text/html;charset=UTF-8");
      try (PrintWriter out = response.getWriter()) {

         out.println("<!DOCTYPE html>");
         out.println("<html>");
         out.println("<head>");
         out.println("<title>Servlet LoginServlet</title>");
         out.println("</head>");
         out.println("<body>");
```

```
            String uname=request.getParameter("txtId");
            String upass=request.getParameter("txtPass");
            if(uname.equals("admin")&&upass.equals("12345"))
            {
             out.println("<body bgcolor=pink>");
             out.println("<h1>Welcome!!! "+uname+"<h1>");
            }
            else
            {
             out.println("<body bgcolor=cyan>");
             out.println("<h1>Login Failed!<h1>");
            }
            out.println("</body>");
            out.println("</html>");
        }
    }
}
```







c) Create a simple calculator application using servlet.

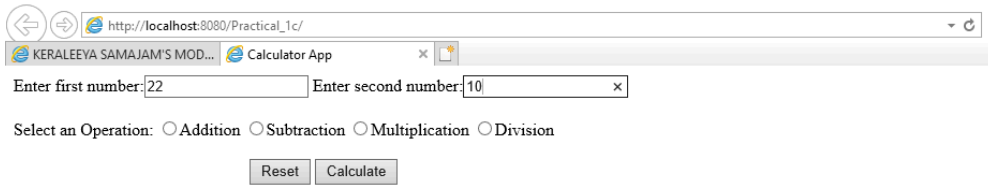
index.html

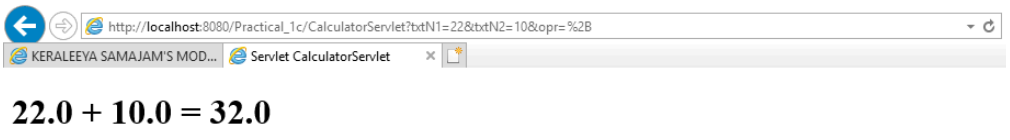
```
<!DOCTYPE html>
<html>
  <head>
    <title>Calculator App</title>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
  </head>
  <body>
    <form method="get" action="CalculatorServlet">
      Enter first number:<input type="text" name="txtN1">
      Enter second number:<input type="text" name="txtN2"><br><br>
      Select an Operation:
      <input type="radio" name="opr" value='+'>Addition
      <input type="radio" name="opr" value='-'>Subtraction
      <input type="radio" name="opr" value='*'>Multiplication
      <input type="radio" name="opr" value='/'>Division<br><br>
          
          
          
          
          
      <input type="reset">
      <input type="submit" value="Calculate">
    </form>
  </body>
```

```
</html>
```

CalculatorServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
public class CalculatorServlet extends HttpServlet {
@Override
   protected void doGet(HttpServletRequest request, HttpServletResponse response)
         throws ServletException, IOException {
      response.setContentType("text/html;charset=UTF-8");
      try (PrintWriter out = response.getWriter()) {
         out.println("<!DOCTYPE html>");
         out.println("<html>");
         out.println("<head>");
         out.println("<title>Servlet CalculatorServlet</title>");
         out.println("</head>");
         out.println("<body>");
         double n1 = Double.parseDouble(request.getParameter("txtN1"));
         double n2 = Double.parseDouble(request.getParameter("txtN2"));
         double result=0;
         String opr = request.getParameter("opr");
         if(opr.equals("+"))
          result=n1+n2;
         if(opr.equals("-"))
          result=n1-n2;
         if(opr.equals("*"))
          result=n1*n2;
         if(opr.equals("/"))
          result=n1/n2;
         out.println(" "+n1+" "+opr+" "+n2+" = "+result);
         out.println("</body>");
         out.println("</html>");
      }
   }
}
```


http://localhost:8080/Practical_1c/
KERALEEYA SAMAJAM'S MOD...
Calculator App
Enter first number: 22
Enter second number: 10
Select an Operation: Addition Subtraction Multiplication Division
Reset
Calculate


http://localhost:8080/Practical_1c/CalculatorServlet?txtN1=22&txtN2=10&opr=%2B
KERALEEYA SAMAJAM'S MOD...
Servlet CalculatorServlet
22.0 + 10.0 = 32.0

Practical 2

a) Create a Registration Servlet in Java using JDBC. Accept the details such as Username, Password, E-mail and Country from the user using HTML form and store the registration details in the database.

index.html

```
<html>
   <head>
      <title>Register</title>
      <meta charset="UTF-8">
      <meta name="viewport" content="width=device-width, initial-scale=1.0">
   </head>
   <body>
      <form action="RegistrationServlet" method="post">
         Username:<input type="text" name="uname"><br>
         Password:<input type="password" name="pw"><br>
         Email Id:<input type="text" name="email"><br>
         Country:<select name="coun">
            <option>Select...
            <option>India
            <option>Bangladesh
            <option>Bhutan
            <option>Canada
         </select><br>
         <input type="submit" value="Register">
      </form>
   </body>
</html>
```

RegistrationServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import java.sql.*;
import java.io.*;
import java.util.*;
import javax.servlet.*;
import javax.servlet.http.*;
```

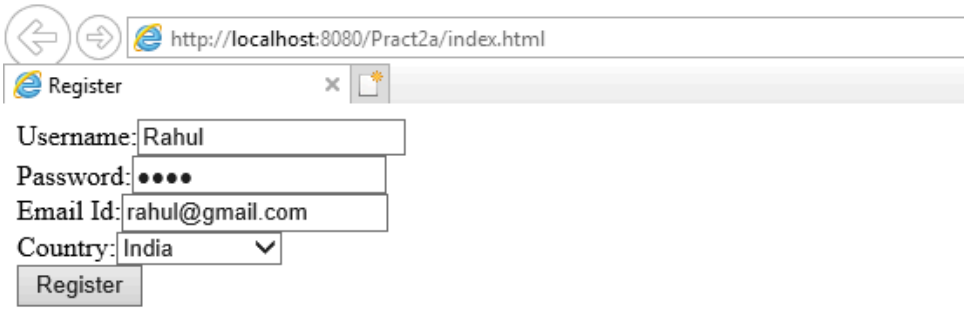
```
public class RegistrationServlet extends HttpServlet {

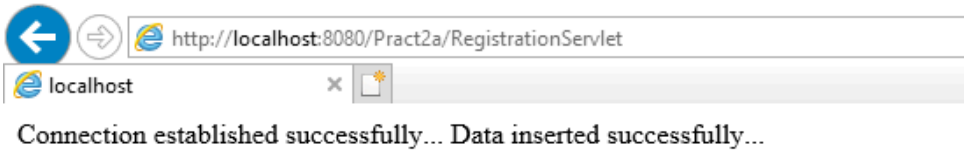
   @Override
   public void doPost(HttpServletRequest req, HttpServletResponse res)
         throws ServletException, IOException {
      Connection con=null;
      PreparedStatement ps=null;
      res.setContentType("text/html;charset=UTF-8");
      PrintWriter out = res.getWriter();
      String username = req.getParameter("uname");
      String password = req.getParameter("pw");
      String emailid = req.getParameter("email");
      String country = req.getParameter("coun");
      try
      {
         Class.forName("com.mysql.jdbc.Driver");

con=DriverManager.getConnection("jdbc:mysql://localhost:3306/mydatabase","root",
"");
         out.println("Connection established successfully...");
         ps= con.prepareStatement("insert into user values(?,?,?,?)");
         ps.setString(1,username);
         ps.setString(2,password);
         ps.setString(3,emailid);
         ps.setString(4,country);
         ps.execute();
         out.println("Data inserted successfully...");
      }
      catch(ClassNotFoundException | SQLException e)
      {
       out.println("<b>" + e + "</b>");
      }
   }
}
```

```
mysql> create database mydatabase;
Query OK, 1 row affected (0.00 sec)

mysql> use mydatabase;
Database changed
mysql> create table user(username varchar(20), password varchar(20), email varchar(20), country varchar(20));
Query OK, 0 rows affected (0.03 sec)
```






```
mysql> select * from user;
+----------+----------+-----------------+---------+
| username | password | email           | country |
+----------+----------+-----------------+---------+
| Rahul    | rnam     | rahul@gmail.com | India   |
+----------+----------+-----------------+---------+
1 row in set (0.00 sec)
```

b) Create a Web Application for user authentication using Servlet and MySQL database.

index.html

```
<!DOCTYPE html>
<html>
  <head>
    <title>User Authentication</title>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
  </head>
```

```
    <body>
        <form method="post" action="UserAuthServlet">
            Enter username: <input type="text" name="txtname">
            Enter password: <input type="password" name="txtpass">`
            <input type="submit" value="Submit">
        </form>
    </body>
</html>
```

UserAuthServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import java.sql.*;
import java.io.*;
import java.util.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class UserAuthServlet extends HttpServlet {
    Connection con=null;
    Statement stmt=null;
    ResultSet rs=null;
    String n="",p="";
    @Override
    public void doPost(HttpServletRequest req, HttpServletResponse res)
            throws ServletException, IOException {
        res.setContentType("text/html;charset=UTF-8");
        PrintWriter out = res.getWriter();
        int cnt=0;
        n=req.getParameter("txtname");
        p=req.getParameter("txtpass");
        try
        {
            Class.forName("com.mysql.jdbc.Driver");

con=DriverManager.getConnection("jdbc:mysql://localhost:3306/userdb","root","");
            stmt=con.createStatement();
            rs=stmt.executeQuery("select * from user");
            while(rs.next())
```

```
            {
              String un=rs.getString(1);
              String upass=rs.getString(2);
              if(n.equals(un)&&p.equals(upass))
              {
                 out.println("<h1> Welcome! " +un+"</h1>");
                 cnt++;
                 break;
              }
            }
            if(cnt==0)
            {
               out.println("<h1>Sorry! Try Again!</h1>");
            }
        }
        catch(ClassNotFoundException | SQLException e)
        {
         out.println(e);
        }
    }
}
```

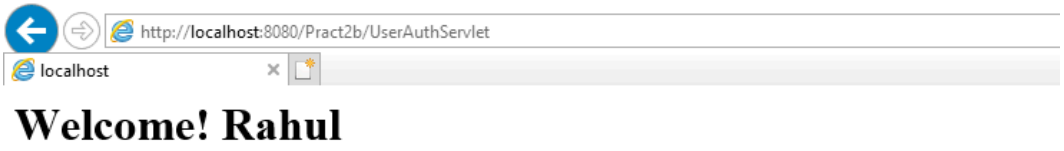
```
mysql> create database userdb;
Query OK, 1 row affected (0.00 sec)

mysql> use userdb;
Database changed
mysql> create table user(uname varchar(20), upassword varchar(20));
Query OK, 0 rows affected (0.02 sec)

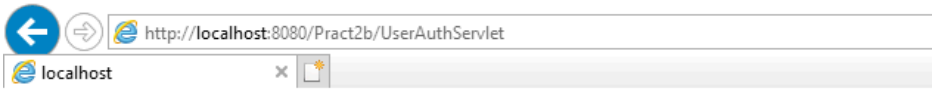
mysql> insert into user values("Rahul","rnam");
Query OK, 1 row affected (0.01 sec)

mysql> select * from user;
+-------+-----------+
| uname | upassword |
+-------+-----------+
| Rahul | rnam      |
+-------+-----------+
1 row in set (0.00 sec)
```





Welcome! Rahul



Sorry! Try Again!

Practical 3

a) Using Request Dispatcher Interface create a Servlet which will validate the password entered by the user, if the user has entered "Servlet" as password, then he will be forwarded to Welcome Servlet else the user will stay on the index.html page and an error message will be displayed.

index.html

```
<html>
	<head>
	<title>Request Dispatcher</title>
	<meta charset="UTF-8">
	<meta name="viewport" content="width=device-width, initial-scale=1.0">
	</head>
	<body>
	<h1> Login Form </h1>
	<form method="post" action="MainServlet">
	Name:       <input type="text"
name="uname"/></br></br>
	Password: <input type="password" name="upass"/></br></br>

              
	   
      <input type="submit" value="LOGIN"/>
	</form>
	</body>
</html>
```

MainServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import javax.servlet.RequestDispatcher;
public class MainServlet extends HttpServlet {
	@Override
	protected void doPost(HttpServletRequest request, HttpServletResponse
response)
	throws ServletException, IOException {
```

```
        response.setContentType("text/html;charset=UTF-8");
            try (PrintWriter out = response.getWriter())
            {
            out.println("<!DOCTYPE html>");
            out.println("<html>");
            out.println("<head>");
            out.println("<title>Servlet MainServlet</title>");
            out.println("</head>");
            out.println("<body>");

            String n=request.getParameter("uname");
            String p=request.getParameter("upass");
            if(n.equals("admin") && p.equals("Servlet"))
            {
                    RequestDispatcher rd=
request.getRequestDispatcher("SecondServlet");
                    rd.forward(request, response);
            }
            else
            {
                    out.print("sorry UserName or Password Error!");
                    RequestDispatcher rd= request.getRequestDispatcher("/index.html");
                    rd.include(request, response);
            }
            out.println("</body>");
            out.println("</html>");
            }
            }
}
```

SecondServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

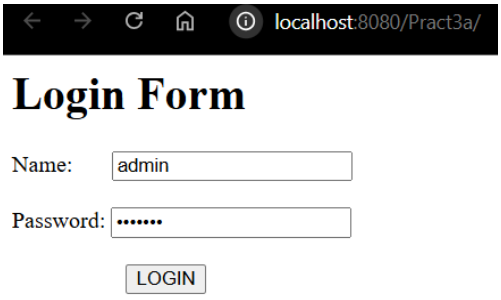
public class SecondServlet extends HttpServlet
{
        @Override
        protected void doPost(HttpServletRequest request, HttpServletResponse
response)
```

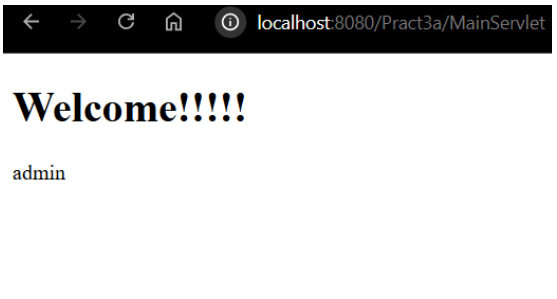
```
        throws ServletException, IOException {
    response.setContentType("text/html;charset=UTF-8");
        try (PrintWriter out = response.getWriter()) {

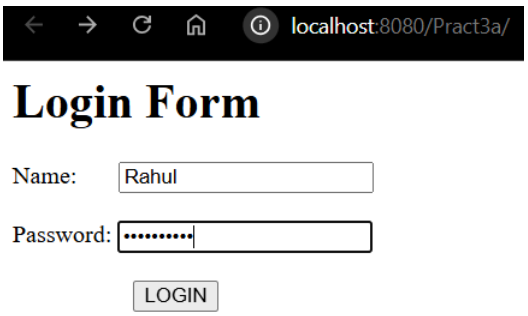
        out.println("<!DOCTYPE html>");
        out.println("<html>");
        out.println("<head>");
        out.println("<title>Servlet SecondServlet</title>");
            out.println("</head>");
        out.println("<body>");

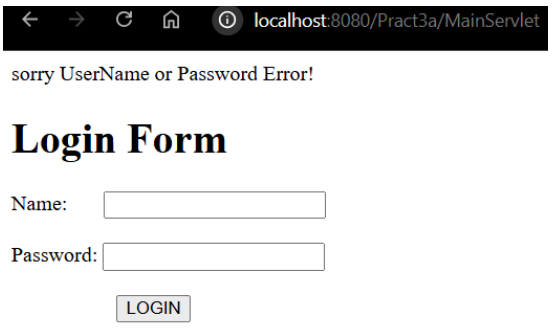
String name=request.getParameter("uname");
        out.println("<h1> Welcome!!!!!</h1>"+name);

        out.println("</body>");
        out.println("</html>");
        }
        }
}
```









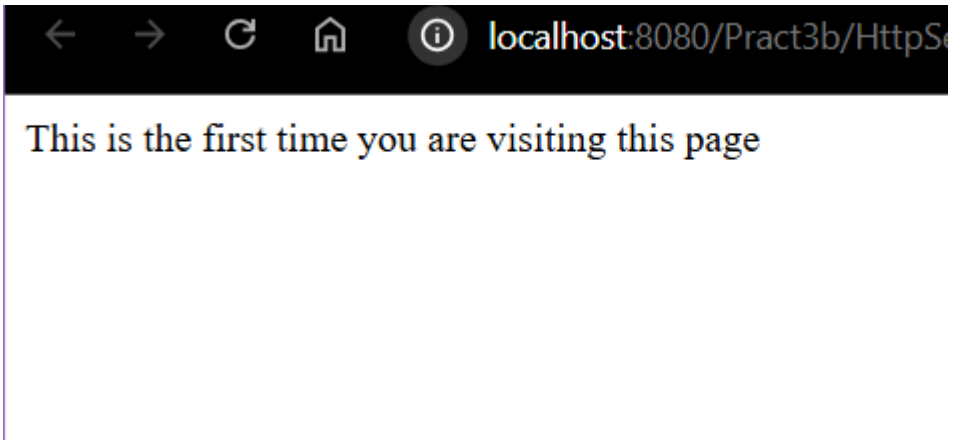
b) Create a servlet demonstrating the use of session creation and destruction. Also check whether the user has visited this page first time or has visited earlier also using sessions.

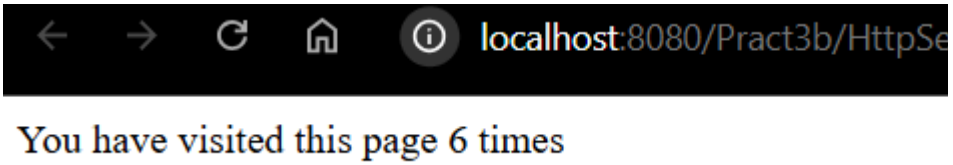
HttpSessionServlet.java

```
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;
public class HttpSessionServlet extends HttpServlet
{
private int counter;
@Override
public void doGet(HttpServletRequest request, HttpServletResponse response) throws
ServletException, IOException
 {
   response.setContentType("text/html");
   PrintWriter out = response.getWriter();
   HttpSession session=request.getSession(true);

   if(session.isNew())
         {
         out.print("This is the first time you are visiting this page");
         ++counter;
         }
         else
         {
               synchronized(HttpSessionServlet.this)
                  {
                     if(counter==10)
                      {
                           session.invalidate();
                           counter=0;
                            request.getSession(false);
                      }
                     else
                            out.print("You have visited this page "+(++counter)+ " times");
                  }
         }
}
```

```
}
```


localhost:8080/Pract3b/HttpS
This is the first time you are visiting this page


localhost:8080/Pract3b/HttpSe
You have visited this page 6 times

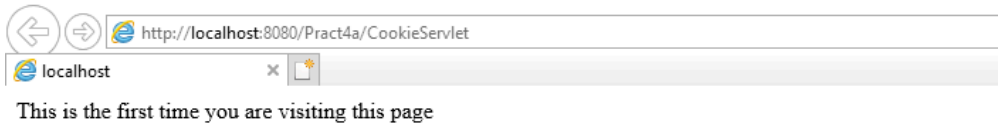
Practical 4

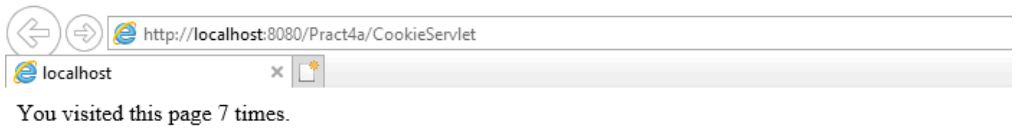
a) Create a Servlet that uses cookies to store the number of times a user visited a Servlet.

CookieServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.Cookie;
import javax.servlet.http.*;

public class CookieServlet extends HttpServlet {
   private int i=1;
   public void doGet(HttpServletRequest req, HttpServletResponse res)
         throws ServletException, IOException {
      res.setContentType("text/html;charset=UTF-8");
      PrintWriter out = res.getWriter();
        String k = String.valueOf(i);
        Cookie c = new Cookie("visit",k);
        res.addCookie(c);
        int j= Integer.parseInt(c.getValue());
        if(j==1)
        {
           out.println("This is the first time you are visiting this page");
        }
        else
        {
         synchronized(CookieServlet.this)
         {
           out.println("You visited this page "+i+" times.");
         }
        }
        i++;
    }
}
```





b) Write a Servlet that accepts a username from an HTML form and stores it as a cookie variable. Write another servlet that returns the value of this cookie variable and displays it.

Cookieshtml.html

```
<!DOCTYPE html>
<html>
   <head>
      <title>CookiesHTML</title>
      <meta charset="UTF-8">
      <meta name="viewport" content="width=device-width, initial-scale=1.0">
   </head>
   <body>
      <form action="AddCookieServlet" method="post">
         Name:<input type="text" name="username"></br>
         <input type="submit" value="Go">
      </form>
   </body>
```

```
</html>
```

AddCookieServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class AddCookieServlet extends HttpServlet {

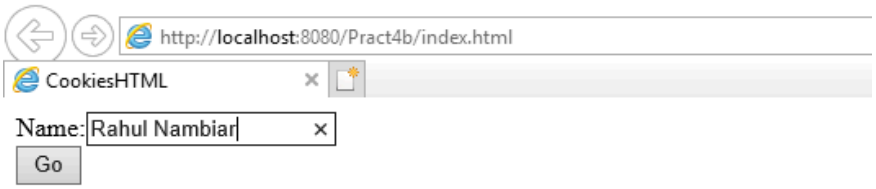
   public void doPost(HttpServletRequest req, HttpServletResponse res)
         throws ServletException, IOException {
       try
       {res.setContentType("text/html;charset=UTF-8");
        PrintWriter out = res.getWriter();
        String n = req.getParameter("username");
        out.println("Welcome "+n);
        Cookie ck = new Cookie("uname",n);
        res.addCookie(ck);
        out.print("<form action='GetCookie' method='post'>");
        out.print("<input type='submit' value='Go'>");
        out.print("</form>");
        out.close();
       }
       catch(Exception e)
       {
         System.out.println(e);
       }
   }
}
```

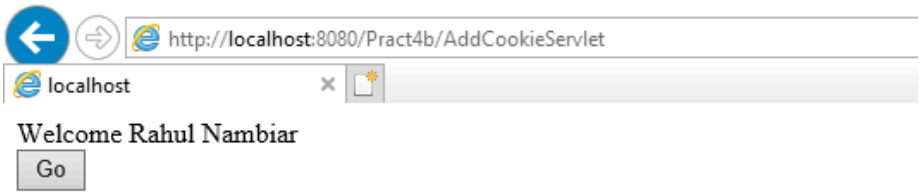
GetCookie.java

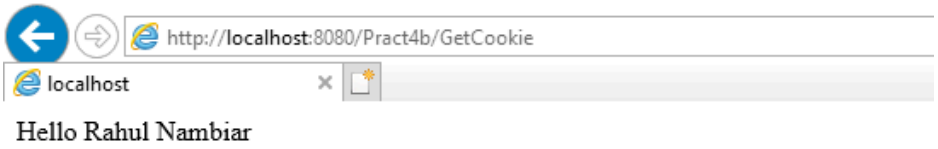
```
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
```

```
import javax.servlet.http.HttpServletResponse;
import java.io.*;
import javax.servlet.*;
import javax.servlet.http.*;

public class GetCookie extends HttpServlet {
  @Override
  public void doPost(HttpServletRequest req, HttpServletResponse res)
        throws ServletException, IOException {
      try
      {
       res.setContentType("text/html;charset=UTF-8");
       PrintWriter out = res.getWriter();
       Cookie ck[] = req.getCookies();
       out.print("Hello "+ck[0].getValue());
       out.close();
      }
      catch (Exception e)
      {
       System.out.println(e);
      }
   }
}
```




http://localhost:8080/Pract4b/AddCookieServlet
localhost
Welcome Rahul Nambiar
Go


http://localhost:8080/Pract4b/GetCookie
localhost
Hello Rahul Nambiar